

7a	When atoms are excited, the electrons move from a lower energy level to a higher energy level. When the <u>electrons de-excite from a higher level to a lower level, they emit photons with energies corresponding to the differences in energy levels of the atoms,</u> giving rise to line emission spectrum. <u>Since the frequencies of the photons are fixed, the photons emitted have discrete amounts of energies. Hence, energy levels are discrete.</u> <u>Examiner's comments:</u> Many students did not explain how the energy of the photon is linked to the energy levels in atoms. A few students described the observations for absorption line spectrum.	A	M1 A1

7b	Since $d \sin \theta = \lambda$ for 1st order maxima, and Hence, $d\left(\frac{L}{D}\right) = 410 \times 10^{-9} \quad (1)$ $d\left(\frac{L + 0.008}{D}\right) = 434 \times 10^{-9} \quad (2)$ From Equation 1 and 2, $L = 0.1367 \text{ m}$ Using, $d\left(\frac{L + 0.075}{D}\right) = X \quad (3)$ From Equation 1 and 3, $X = 635 \times 10^{-9} \text{ m}$ Or Using idea of $d \sin \theta = \lambda$ and small angle approximation, $\theta \propto \lambda$ Since $\theta \propto \lambda$, $\Delta \theta \propto \Delta \lambda$, $\Delta L \propto \Delta \lambda$ $\frac{434 - 410}{0.8} = \frac{X - 410}{8.2}$ $X = 656 \text{ nm}$	D M1 A1 (M1) (A1)
7c	Energy of photon with wavelength 410 nm $= \frac{hc}{\lambda}$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{410 \times 10^{-9}}$ $= 4.85 \times 10^{-19} \text{ J}$ Energy level = $4.85 \times 10^{-19} + (-4.08 \times 10^{-19})$ $= 0.77 \times 10^{-19} \text{ J}$	D M1 M1

			A1
7d	The lowest energy level shown in Fig 7.3 may not be the energy level at ground state.	D	A1
7e	When the electrons are directed to pass through a <u>thin carbon film</u>, <u>concentric rings</u> are observed It shows a <u>diffraction pattern</u> that is unique to wave behaviour after passing through the carbon film.	A	M1 A1
7f	$E_k \text{ of electron} = \frac{p^2}{2m}$ $p = \sqrt{2mE_k}$ By de Broglie's equation, $\lambda = \frac{h}{p}$ $= \frac{h}{\sqrt{2mE_k}}$ $= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(4.08 \times 10^{-19})}}$ $= 7.69 \times 10^{-10} \text{ m}$	A	C1 A1

8a	greater lattice vibrations	A	B1
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